

Part D

1. Which model is best between Model 1 and Model 2?

The model that did the best was the Semantic Search Model based on the following evaluation metrics Model 2 Semantic Search Precision@10 95.65 Recall@10 : 5.17 NDCG@10 : 88.46 . However besides the evaluation metrics , when we looked at the type of model Semantic Search is and how I have defined it in Part B of the project we see how “user friendly” and intuitive it is. The model is a neural network model and therefore is able to learn and make more complex connections

This is further illustrated in the practical example of the “boho bed frame” and how in the BM25 the model was ranking picture frames , pillow cases and curtain frame in the top ten but the most relevant product which was title boho chic bed platform the BM25 model rant 26th place while the same query and product pair ranked first in the Semantic Search Model.

2. How can the selected model be improved?

The current model was a MVP to prove to Stadioalot executives that the project and problem statement is real and can be solved and therefore we used external datasets from oversea companies but for an improved implementation we would have to understand Stadioalot's specific customer and their specific queries instead of more general queries , process is called Fine-tuning. According to Roussinov, Sharoff and Puchnina (2023) fine-tuning is the machine learning process of taking pre-trained general language model and continually training it on a smaller task specific dataset, the pre-training gives context on grammar and syntax while fine-tuning identifies semantic relationships by adapting neural weights . Furthermore, we could also use customer behaviour to further fine-tune the model for example, if a customer repeatedly searches for the same brand, that brand might rank higher.

3. How will the selected model need to be adapted to work with your requested data from SS1?

In SS1 I stated that I required 2 sets of data , namely a product catalogue and Sellers data then secondly I would need Customer Search Query and Behavioural Data. The WANDA dataset has 3 sets of Data, a product catalogue , customer search queries and lastly a label of the product search query pair which was judged by humans. Therefore I think I was largely correct with my required information for the Stadiolot Information Retrieval Models. However the product catalogue would have to be cleaned and

combined to create the product document and a sentence transformer would have to embed the product document . The same process embedding would have to be done for the customer search queries. However behavioural data like clicks , add to cart and ratings could have the same effect as the label data had to help compare rankings and give a richer detail on the products ranking.

4. Does your results align with the related literature from Part A?

This is an interesting question because 2/3 of the articles that were profiled in Part A used the BM25 model, this tracks because BM25 is more of a traditional information retrieval model. But I believe that the experiment of the boho bed frame shows the weakness of BM25 ,in its ability to contextual natural language and to rank . If we were to look back at the evaluation metrics the two models scored almost identically however Semantic Search is better able to " learn" and contextualize rather than compare word for word and that my research statement hope to look at , understanding the commercial viability of the model to increase sales and Semantic Search model is better able to understand under intent.

Reference List Entry:

Roussinov, D., Sharoff, S. and Puchnina, N. (2023) 'Fine-tuning language models to recognize semantic relations', *Language Resources and Evaluation*, 57, pp. 1463–1486. doi:10.1007/s10579-023-09641-z.